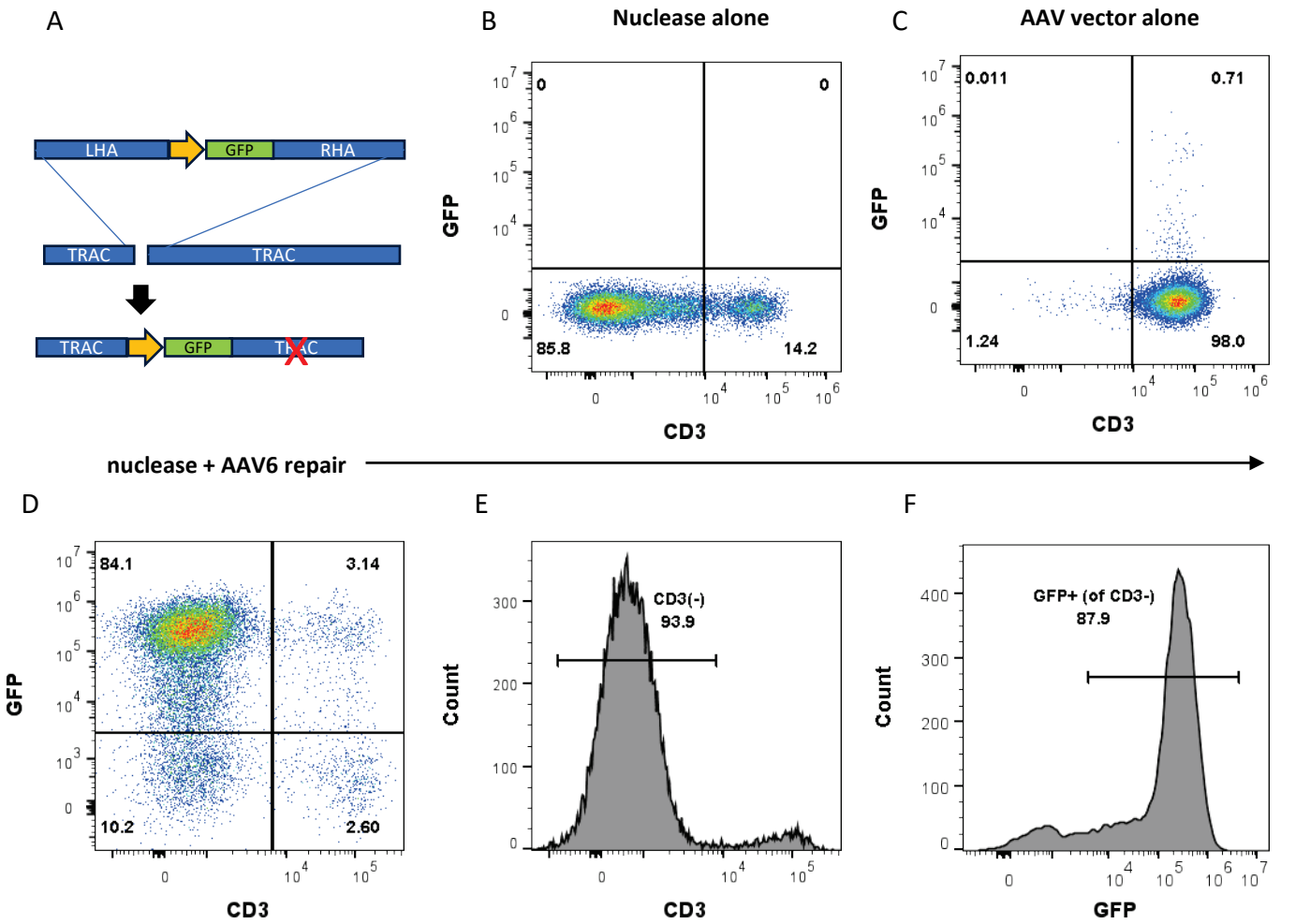
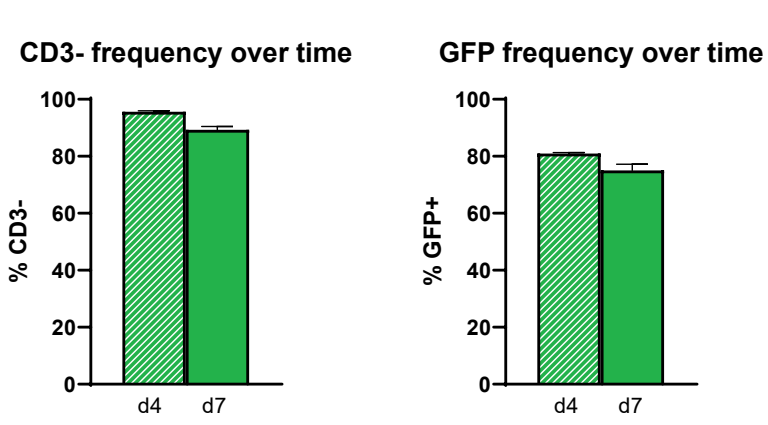
Supplementary Figures

Supplementary Figure 1

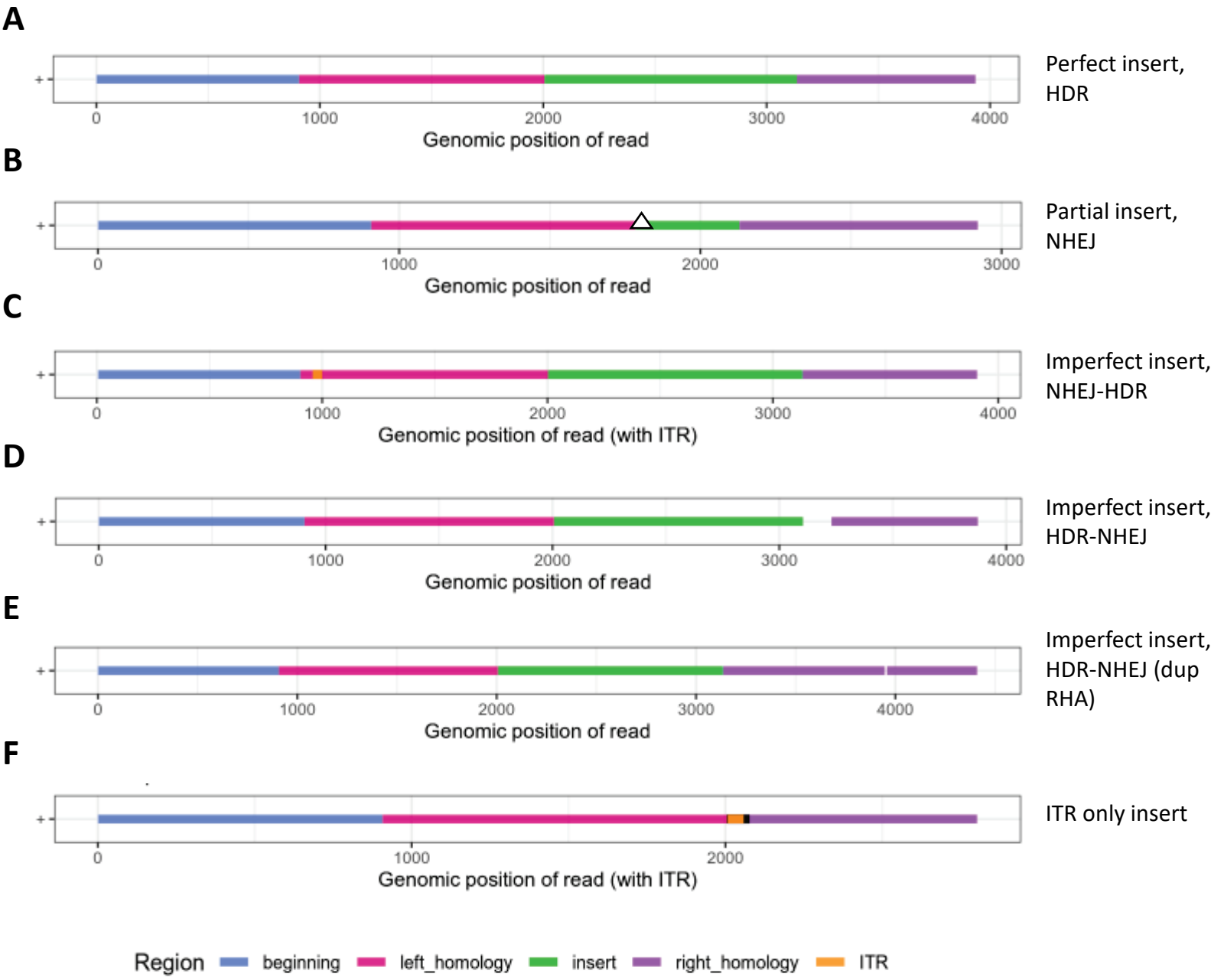




Supplementary Figure 1 legend

High efficiency gene insertion achieved in human T cells. Three days after activation, human T cells were either mock electroporated, or electroporated with ARCUS mRNA (500ng/1e6 cells). After electroporation, cells were either mock transduced or transduced with AAVs at $4.0e^4$ MOI and expanded until 7 days post electroporation, when they were analyzed by flow cytometry for GFP, and CD3. The insert strategy and resulting allele are shown in panel A. GFP and CD3 gates are drawn based on T cells that were electroporated and treated with (B) nuclease alone, (C) mock-electroporated and infected with AAV vector only or (D) electroporated and treated with AAV6 repair vector. Panel E displays the overall frequency of *TRAC*-edited (CD3-) cells while panel F (gated on CD3-) displays the frequency of cells that contain a transgene (GFP+). The frequency of edited cells (CD3-) and insert-bearing cells (GFP+) was determined at d4 and d7 following editing and the results are shown in panel G. Experiments are representative of 3 biological replicates each containing 3 technical replicates using 2 different human donors.

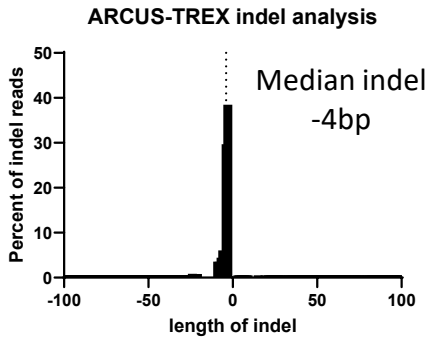
Supplementary Figure 2



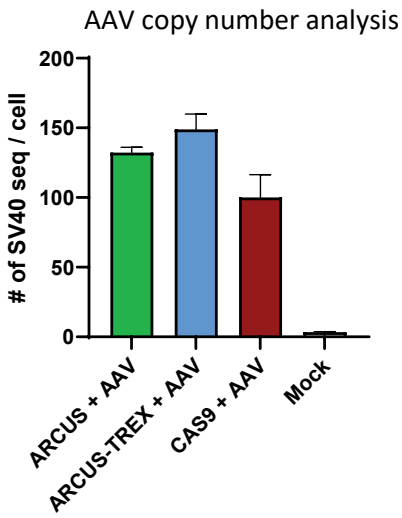
Supplementary Figure 2. Depicted above are the six most common long-read sequencing outcomes for insert-positive repaired alleles analyzed in Figure 1I. Inserts with seamless integration on both ends are the most common outcome detected by NanoPore long-read sequencing (A) followed by the less common errors in gene insertion (B-F).

Supplementary Figure 3

A



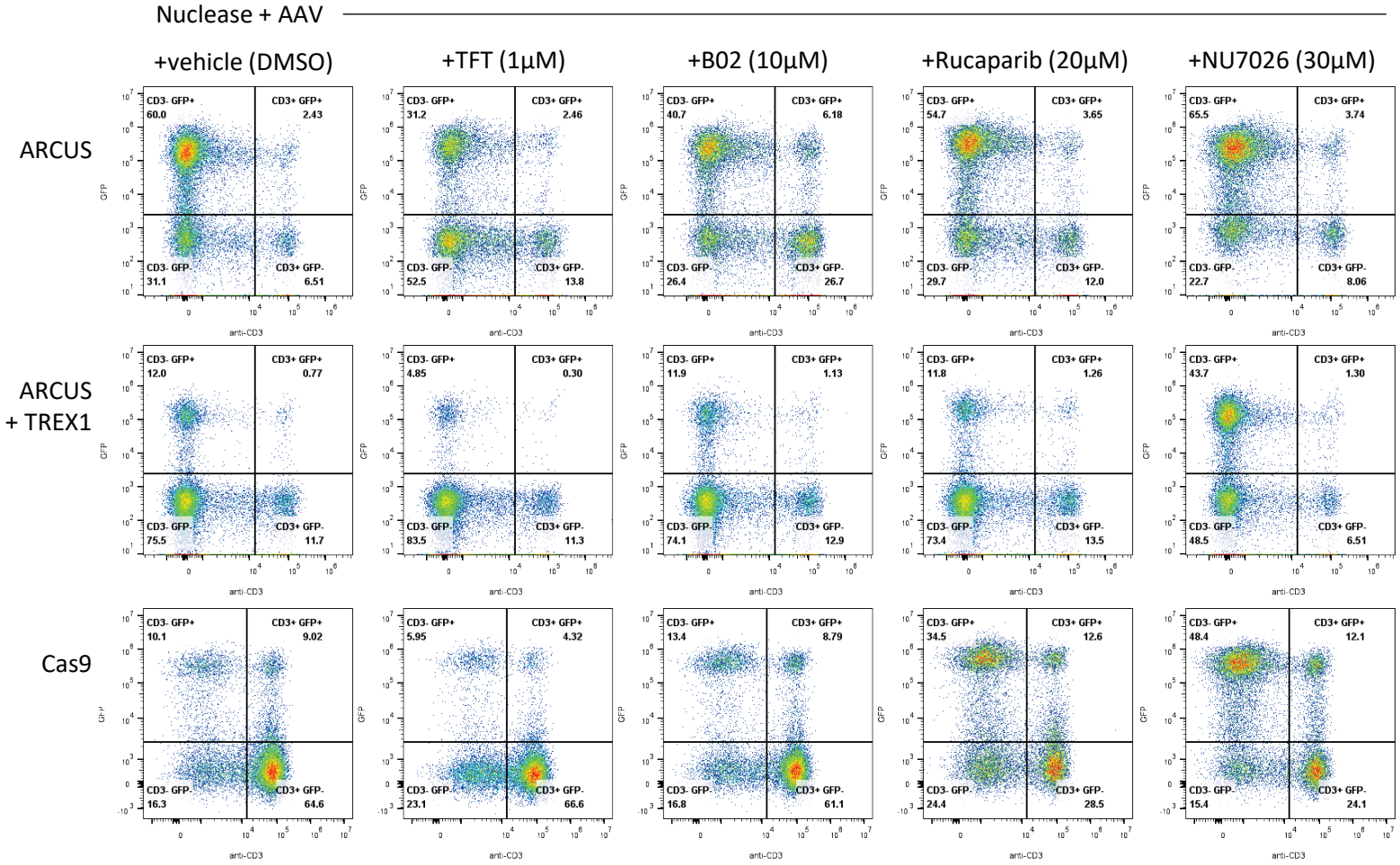
B



Supplementary Fig 3 Legend

ARCUS-TREX1 fusion alters the DSB and does not reduce repair template abundance. We analyzed the size and composition of insertions and deletions in the *TRAC* locus following gene editing. T cells were electroporated with optimized amounts of mRNA encoding ARCUS-TREX in the absence of repair template. Seven days later, genomic DNA was collected and indels were analyzed via AmpSeq. The distribution in indel size is shown in panel A where the vertical dotted line denotes the median indel size of -4bp. Data are representative of two biological replicates. In panel B, genomic DNA was collected 24h following TRAC editing and AAV transduction. AAV copy number was measured by a ddPCR assay targeting the SV40 polyA signal sequence which terminates GFP transcription in the transgene.

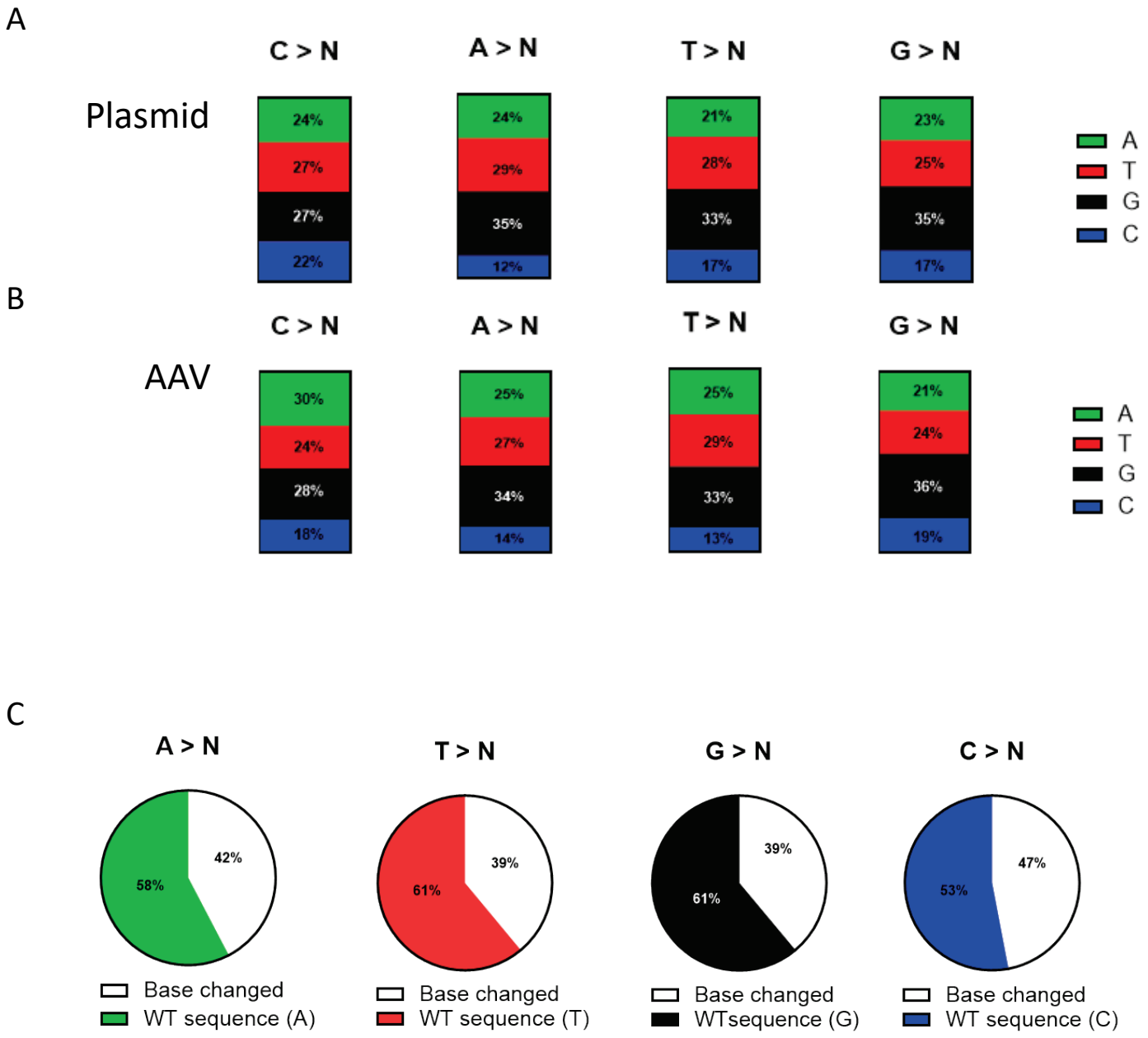
Supplemental Figure 4



Supplementary Fig 4 Legend

DDR pathway inhibitors exert distinct effects on staggered and blunt DSBs. Pictured above are representative dot plots supporting Figure 4. T cells were edited at the TRAC locus with ARCUS (top row), ATCUS-TREX (middle row) or CRISPR-Cas9 (bottom row) and transduced with repair template in the presence of the indicated small molecule. Seven days later, cells were stained for CD3 and analyzed for CD3/TCR display and GFP expression.

Supplementary Figure 5



Supplementary Figure 5 Legend

AAV packaging is accomplished with minimal library bias. T cells were electroporated with mRNA encoding an ARCUS nuclease specific for *TGFBR2* and transduced with one of four AAV repair vectors with degeneracy at the positions indicated in Figure 6. NGS analyses of the plasmid used to produce the AAV and the AAV genomes are shown in panels A and B respectively. The base composition at the four positions (left to right) are shown as percentages in the stacked bar graphs. The pie charts in panel C show the overall frequency of base changes at the indicated positions shown in white. Data are representative of three biological replicates.

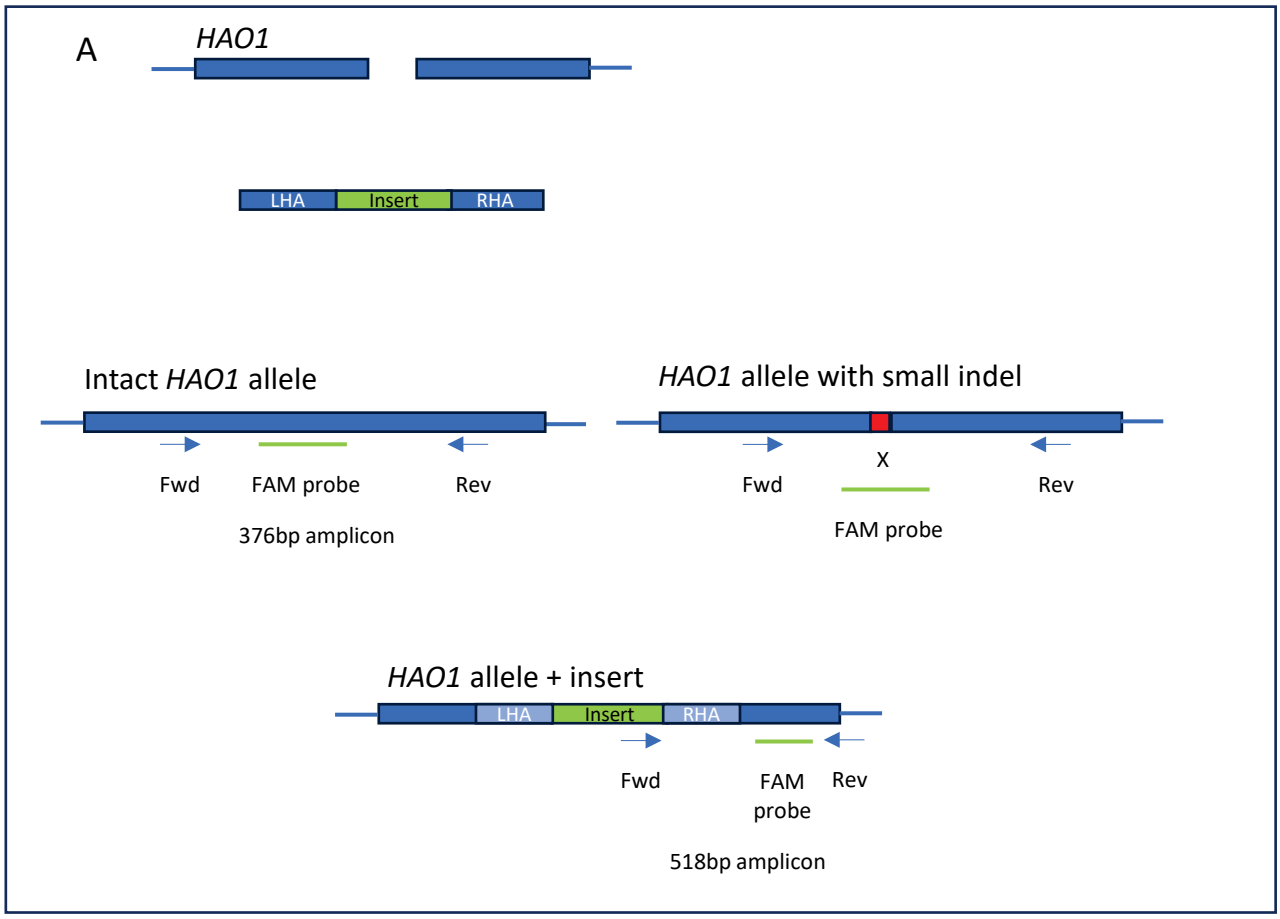
Supplementary Table 1

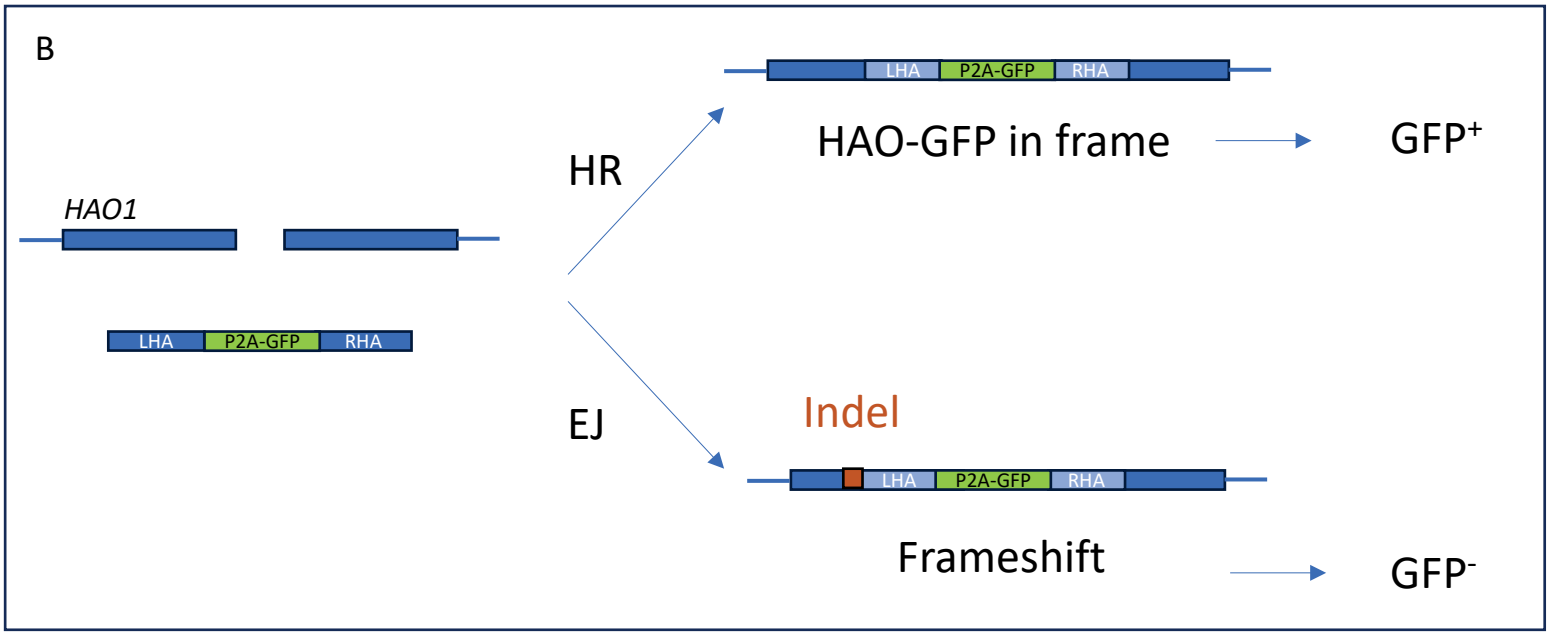
Homology Arm Length Comparison

Homology Arm Length	Insert Frequency (%)
500bp	78.65%
200bp	79.31%
50bp	65.14%

HDR is effectively supported with homology arms between 50 and 500bp in length. T cells were electroporated with mRNA encoding an ARCUS nuclease directed against the TGFBR2 locus and transduced with one of three AAV6 vectors encoding TGFBR2-specific repair templates. The repair templates encoded a JeT-GFP construct flanked by 50, 200, or 500bp homology arms. Seven days later, the cells were analyzed for GFP frequency and the results appear in the table above.

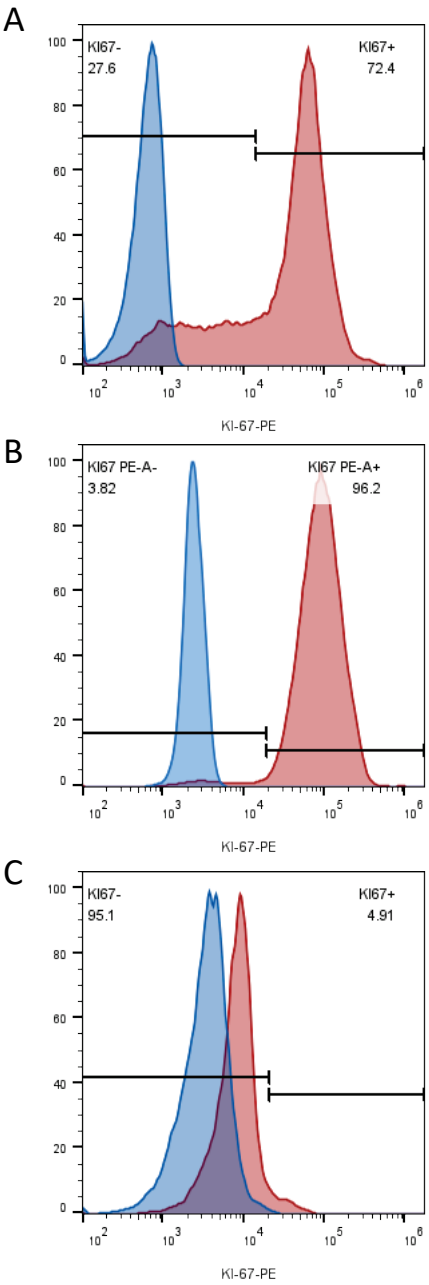
Supplementary Figure 6





Schematic representation of insert detection strategies in primary human hepatocytes: Primer and probe locations are shown for *HAO1* and homologous repair vectors encoding a ddPCR primer binding site (A). Detection of intact and disrupted *HAO1* alleles as well as an insert-positive allele are shown. Functioning of the promoterless GFP vector is depicted in panel B.

Supplementary Figure 7



Primary human hepatocytes are nondividing. T cells stimulated for 3d with anti-CD3/CD28/CD2 (A), AML14 myeloid leukemia cells in culture (B), and primary human hepatocytes (C) were formalin-fixed, permeabilized, and analyzed for expression of the cell proliferation marker Ki-67 by flow cytometry. Blue histogram curves represent unstained negative controls while red curves were stained with anti-Ki-67-PE.

Supplementary Table 2: Software packages used for NGS analyses

Tool Name	Version	Description	License type	PMID	Repository URL
long-amp-nf	NA	custom code for analyzing long-amplicons for HDR/NHEJ insert	NA	NA	Provided as supplementary file.
Samtools	v1.16.1	SAMtools is a widely used program for processing and analysing high-throughput sequencing data.	MIT/Expat License	33590861	https://github.com/samtools/samtools
Seqkit	v2.8.2	a cross-platform and ultrafast toolkit for FASTA/Q file manipulation	MIT License	38898985	https://bioinf.shenwei.me/seqkit/
NanoStat	1.6.0	Calculate various statistics from a long read sequencing dataset in fastq, bam or albacore sequencing summary format.	GPL-3.0 license	29547981	https://github.com/wdecoster/nanostat
porechop	0.2.4	Porechop is a tool for finding and removing adapters from Oxford Nanopore reads	GPL-3.0 license	NA	https://github.com/rrwick/Porechop
pychopper	2.7.10	Pychopper v2 is a tool to identify, orient and trim full-length Nanopore cDNA reads. The tool is also able to rescue fused reads	Oxford Nanopore Technologies PLC. Public License Version 1.0: https://github.com/epi2me-labs/pychopper?tab=License-1-ov-file#readme	NA	https://github.com/epi2me-labs/pychopper
ngmlr	0.2.7	Long read aligner	MIT License	29713083	https://github.com/philres/ngmlr
GenomicAlignments	1.38.0	Provides efficient containers for storing and manipulating short genomic alignments (typically obtained by aligning short reads to a reference genome). This includes read counting, computing the coverage, junction detection, and working with the nucleotide content of the alignments.	Artistic-2.0	23950696	https://github.com/Bioconductor/GenomicAlignments
plyranges	1.22.0	A dplyr-like interface for interacting with the common Bioconductor classes Ranges and GenomicRanges. By providing a grammatical and consistent way of manipulating these classes their accessiblity for new Bioconductor users is hopefully increased.	Artistic-2.0	30609939	https://www.bioconductor.org/packages/release/bioc/html/plyranges.html

Supplementary Table 3: Amino acid sequences of I-CreI and engineered ARCUS nucleases

Endonuclease	Amino acid sequence	% homology to I-creI
I-CreI	MNTKYNKEFLLYLAGFVDGDGSIIAQIKPNQSYKFKHQLSLTFQVTQKTQRRWFLDKLVDEIGVGYVRDRGSVSDYILSEIKPLHNFLTQLQPFLKLKQKQANLVLKIIEQLPSAKESPDKFLEVCTWVDQIAALNDSKTRKTTSETVRAVLDSLSEKKKSSP	100%
TRC1	MNTKYNKEFLLYLAGFVDGDGSIYAVIYPHQRAKFKHFLKLLFTVSQSTKRRWFLDKLVDEIGVGYVYDLPRTSEYRLSEIKPLHNFLTQLQPFLKLKQKQANLVLKIIEQLPSAKESPDKFLEVCTWVDQIAALNDSRTRKTTSETVRAVLDSL	83%
TRC2	YNKEFLLYLAGFVDGDGSIYACIRPRQGSKFKHRLTLGFAVGQKTQRRWFLDKLVDEIGVGYVYDRGSVSEYVLSEIKPLHNFLTQLQPFLKLKQKQANLVLKIIEQLPSAKESPDKFLEVCTWVDQIAALNDSKTRKTTSETVRAVLDSLSEKKKSSP	88%
TGF21	MNTKYNKEFLLYLAGFVDADGSIFAQIIPMQDRKFKHRLRLGFRVIQKTQRRWFLDKLVDEIGVGYVYDTGSVSYYALSQIKPLHNFLTQLQPFLKLKQKQANLVLKIIEQLPSAKESPDKFLEVCTWVDQIAALNDSKTRKTTSETVRAVLDSL	85%
TGF22	YNKEFLLYLAGFVDGDGSIFATINPRQRWKFKHQLRLNFQVTQKTQRRWFLDKLVDEIGVGYVRDTGSVSQYYLSEIKPLHNFLTQLQPFLKLKQKQANLVLKIIEQLPSAKESPDKFLEVCTWVDQIAALNDSKTRKTTSETVRAVLDSLSEKKKSSP	91%
HAO25	MNTKYNKEFLLYLAGFVDADGSIWAYIEPCQWVKFKHRLKLQLNVTQKTQRRWFLDKLVDEIGVGYVRDTGSVSQYMLSEIKPLHNFLTQLQPFLKLKQKQANLVLKIIEQLPSAKESPDKFLEVCTWVDQIAALNDSKTRKTTSETVRAVLDSL	86%
HAO26	YNKEFLLYLAGFVDGDGSIYAKIRPQQASKFKHVLELVFEVTQSTQRRWFLDKLVDEIGVGYVYDWKQASMYRLSQIKPLHNFLTQLQPFLKLKQKQANLVLKIIEQLPSAKESPDKFLEVCTWVDQIAALNDSKTRKTTSETVRAVLDSLSEKKKSSP	87%

Supplementary Table 4: 22 base-pair DNA sequences recognized by *TRAC*, *TFGBR2*, and *HAO1*-specific ARCUS nucleases

Locus	22bp ARCUS binding site
TRAC	TGGCCTGGAGCAACAAATCTGA
TGFBR2	CGGACTCCTGTGCAGCTTCCCT
HAO1	ATGGAAGCTGTATCCAAGGATG